Progress Report

Image Captioning using Retrieval-Augmented Generation

Phase 1: Dataset Exploration, Preparation, and Knowledge-Base Foundations

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Project: Image Captioning using Retrieval-Augmented Generation

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Dataset: Microsoft COCO 2017

Development environment: Local machine

Source repository:

https://github.com/ahmedhouas/Image-Captioning-RAG

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Chapter 1

Introduction

Image captioning is a computer vision and natural language processing task that aims to generate a natural-language description of the visual content of an image. An image-captioning system must identify relevant objects and activities, interpret the scene, and express the information in a concise and coherent sentence.

Recent vision-language models have improved the quality of automatically generated descriptions. Nevertheless, the generated captions may omit relevant objects, misinterpret visual relationships, or introduce details that are not sufficiently supported by the image. These limitations motivate the investigation of methods that can provide a generation model with relevant external information.

Retrieval-Augmented Generation (RAG) is an approach in which information is retrieved from an external collection and made available to a generative model as additional context. In an image-captioning application, the general objective is to use relevant information associated with previously observed images and their descriptions to support caption generation.

The project, entitled *Image Captioning using Retrieval-Augmented Generation*, investigates the use of retrieval-supported information for automated image description. The overall research objective is to study whether the use of an external collection of image-caption data can improve the relevance and factual grounding of generated captions.

This document reports on **Phase 1** of the project. It focuses on the dataset, data exploration, preprocessing, embedding preparation, knowledge-base organization, and quality-control activities completed during this phase. It does not report results from a complete retrieval-and-generation system or from later experimental phases.

The work described in this report was implemented on a **local machine**. The dataset and project files were managed through the local development environment and the project repository.

Chapter 2

Problem Statement

2.1 Limitations of standalone image captioning

A conventional image-captioning model generates a description from the visual representation of an input image and the knowledge encoded in its parameters. Although this can produce fluent sentences, fluency does not necessarily guarantee factual accuracy.

The principal limitations motivating this project are:

- **Hallucination:** a model may mention an object, action, or attribute that is not present in the image.
- **Object omission:** relevant objects may be ignored, particularly in crowded or visually complex scenes.
- **Imprecise descriptions:** the model may identify an object correctly but describe its colour, state, position, or interaction inaccurately.
- **Limited specific knowledge:** a model may have difficulty describing uncommon entities or concepts that require more specific contextual information.
- **Insufficient grounding:** a caption may be linguistically plausible without being fully supported by the visual content.

For example, an image showing a person standing beside a bicycle could be described incorrectly as showing a person riding the bicycle. The generated sentence is plausible in a general context, but the action is not necessarily supported by the image.

2.2 Motivation for retrieval-augmented image captioning

RAG provides a framework for supplementing a model's input with information retrieved from an external knowledge source. For image captioning, this project uses image-caption data as the basis for constructing an organized collection of visual and textual representations.

The purpose of Phase 1 is to prepare this data foundation carefully. This requires selecting and separating the dataset, understanding its statistical characteristics, preparing image and caption records, generating their embeddings, and checking the integrity of the resulting artifacts.

The research question guiding the overall project is:

> *To what extent can retrieval-supported information contribute to the relevance and factual grounding of automatically generated image captions?*

Phase 1 establishes the data and knowledge-base foundations needed to investigate this question in the subsequent work.

Chapter 3

Phase 1: Workflow and Project Organization

3.1 Scope of the current phase

Phase 1 concentrates on preparing a reliable and reproducible dataset for the project. The work is organized into the following activities:

1. Inspect the source dataset and its annotations.
2. Define reproducible training, validation, and test splits.
3. Explore the visual and linguistic characteristics of the selected data.
4. Apply the required preprocessing operations.
5. Generate image and caption embeddings.
6. Organize the records, embeddings, and metadata into a knowledge-base structure.
7. Perform integrity checks and a data-leakage audit.
8. Verify that the prepared artifacts can be loaded and used by the current retrieval-index components.

The phase is considered a data-preparation and infrastructure stage. The retrieval compatibility check described later is a technical validation of the prepared artifacts; it is not a claim that a complete caption-generation pipeline has been implemented.

3.2 Local development environment and data management

All implementation and data-management activities reported in this phase were carried out on a local machine. Google Colab and Google Drive were not used as the execution or primary data-management environment.

The project uses a Python-based workflow organized around scripts, configuration files, and Jupyter notebooks. This organization supports modular execution, inspection of intermediate outputs, and reproducibility.

The principal technologies used include:

Table 3.1: Main tools and their roles in Phase 1.

Technology	**Role**
Python	Main programming language for data processing and pipeline scripts.
Jupyter Notebook	Interactive execution, inspection, and documentation of experiments.
Pandas	Tabular manipulation of image and caption annotations.
NumPy	Numerical operations and embedding-array handling.
Matplotlib and Seaborn	Statistical visualization.
spaCy	Natural-language processing and Part-of-Speech analysis.
Pillow	Image loading and image preprocessing.
PyTorch and Transformers	Execution of the CLIP model and associated preprocessing.
Sentence-Transformers	Generation of sentence-level text embeddings.
FAISS	Construction and validation of vector indexes.

The project repository contains the source code, notebooks, configuration, reports, and generated metadata needed to document the work. The original dataset is managed separately from source-code versioning so that large dataset files do not need to be committed to the repository.

3.3 Embedding and knowledge-base preparation

Two embedding models are used to represent the image and caption collections. Their outputs are stored and indexed separately.

3.3.1 Image embeddings using CLIP

The image representation is generated using the pretrained model `openai/clip-vit-base-patch32`. The model produces a 512-dimensional vector for each image:

$$\mathbf{e}_I = f_{\text{CLIP}}(I),$$

where I is an image and f_{CLIP} denotes the image encoder.

The current training collection contains 9,900 image embeddings. The vectors are stored as `float32` values and are L2-normalized before indexing.

3.3.2 Caption embeddings using Sentence-BERT

Textual representations are generated using the Sentence-Transformers model `all-MiniLM-L6-v2`. Each caption is encoded as a 384-dimensional vector:

$$\mathbf{e}_T = f_{\text{SBERT}}(T),$$

where T is a caption and f_{SBERT} is the sentence-embedding model.

The current training collection contains 49,500 caption embeddings. These vectors are also stored as `float32` values and L2-normalized.

3.3.3 Separate embedding spaces

The image and text representations have different dimensions and are produced by different encoders. CLIP image embeddings have 512 dimensions, whereas Sentence-BERT embeddings have 384 dimensions. They are therefore maintained as separate embedding collections and are not directly compared as vectors in a single shared space.

For normalized vectors, inner-product similarity is equivalent to cosine similarity:

$$\text{cosine}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}.$$

The normalization allows the FAISS indexes to use inner-product search for similarity ranking within each respective embedding space.

3.4 Vector indexes and document organization

FAISS is used to create two independent exact-search indexes based on `IndexFlatIP`.

Table 3.2: Embedding collections prepared for the training knowledge base.

Collection	**Vectors**	**Dimensions**	**Encoder**
Image collection	9,900	512	CLIP ViT-B/32
Caption collection	49,500	384	all-MiniLM-L6-v2
Total	59,400	–	–

The associated document records preserve the identifiers and metadata required to map an indexed vector back to its source record.

Image documents include information such as the image ID, filename, embedding row, and associated caption identifiers. Caption documents include the caption ID, associated image ID, image filename, caption text, and embedding row.

This organization preserves the relationship between each image and its five reference captions and allows the embedding collections to be checked against their corresponding metadata.

Chapter 4

Dataset and Splitting Strategy

4.1 Dataset source

The project uses the Microsoft COCO 2017 dataset (Common Objects in Context). COCO provides images together with human-written captions and instance-level object annotations.

The principal source files used in the project are:

- `train2017`: training images.
- `val2017`: images used for the test subset in the current project configuration.
- COCO caption annotations: image-caption associations.
- COCO instance annotations: object categories and instance-level information.

Each selected image is associated with five human-written captions. The instance annotations provide object-category information used in the statistical exploration.

4.2 Dataset configuration

The current dataset configuration contains 10,100 images in total, divided into a knowledge-base training subset, a validation subset, and a test subset.

Table 4.1: Dataset split configuration.

Split	**Images**	**Captions**	**Source**
Training / knowledge base	9,900	49,500	COCO train2017
Validation	100	500	COCO train2017
Test	100	500	COCO val2017
Total	10,100	50,500	–

The 10,000 selected images from `train2017` are divided into 9,900 training images and 100 validation images. The 100 test images are selected separately from `val2017`.

4.3 Configurable and reproducible splitting strategy

The project implements a configurable split-generation mechanism rather than relying on manually maintained image lists. The split sizes and random seed are centralized in `01_src/config/pipeline_config.py`. The current configuration uses:

- `RANDOM_SEED = 45`;
- `TRAINING_SIZE = 9900`;
- `VALIDATION_SIZE = 100`;
- `TEST_SIZE = 100`;
- `TOTAL_TRAIN_POOL = TRAINING_SIZE + VALIDATION_SIZE`;
- `CAPTIONS_PER_IMAGE = 5`.

The split-generation logic is implemented in `01_src/data/split_dataset.py`. It reads the configured values and constructs the subsets from the source image identifiers. The 9,900-image training subset and the 100-image validation subset are sampled from the COCO `train2017` pool. The 100-image test subset is selected independently from COCO `val2017`. A fixed seed makes a given configuration reproducible, while changing the configured sizes or seed allows a new split to be generated without manually editing the identifier lists.

The current procedure follows these steps:

1. Read the split configuration and source image identifiers.
2. Sample a pool of `TRAINING_SIZE + VALIDATION_SIZE` identifiers from `train2017`.
3. Shuffle the sampled pool using `RANDOM_SEED`.
4. Allocate the configured number of identifiers to training and validation.
5. Independently sample the configured test size from `val2017`.
6. Validate identifier uniqueness and split separation.
7. Save the resulting manifests and configuration under `02_processed_data/splits/`.

The generated split artifacts are `training_ids.json`, `validation_ids.json`, `test_ids.json`, and `split_config.json`. These files record the actual membership of each subset. Consequently, changing the configuration alone does not retroactively change an already generated split: the split-generation step must be run again to produce updated manifests.

4.3.1 How the pipeline is rerun after a split change

A new split changes which images and captions belong to each subset. All downstream outputs that depend on split membership must therefore be regenerated. The project provides a pipeline entry point, `python 01_src/run_pipeline.py`, to rerun the processing workflow using the current configuration. The corresponding stages are:

1. regenerate the split manifests;
2. extract and validate the annotations for the new image identifiers;
3. rebuild the processed image and caption records;
4. regenerate the image and caption embeddings;
5. rebuild the knowledge-base documents and their FAISS indexes;
6. rerun quality-control checks, split-overlap checks, and retrieval compatibility validation.

The workflow can also be executed through the project's numbered notebooks and modules, in their documented order (from workspace setup and dataset audit through splitting, preprocessing, embedding/index construction, and quality control). The important reproducibility rule is that a changed split must be treated as a new dataset build: the new manifests and all dependent artifacts must belong to the same run. Old embeddings or indexes must not be combined with newly generated split files, because doing so could create inconsistent image-caption mappings or unintended data leakage.

The current configuration yields 9,900 training images, 100 validation images, and 100 independent test images. Validation examples are kept out of the training knowledge base and are handled as validation queries during the retrieval checks. Test images originate from the separate COCO `val2017` source and are not included in the training knowledge base.

Chapter 5

Data Exploration

5.1 Purpose of the exploration

Data exploration is a central component of Phase 1. Before preparing embeddings and indexes, it is necessary to understand the visual and linguistic characteristics of the selected dataset.

The exploration addresses the following questions:

- Which object categories occur most frequently?
- How many annotated objects are present in a typical image?
- How long are the human-written captions?
- How large and diverse is the caption vocabulary?
- Which grammatical categories are most common?
- How frequently do captions mention colours, shapes, and scene contexts?
- Which object categories commonly occur in the same image?

The statistics discussed in this chapter concern the 9,900-image training subset and its 49,500 captions, unless otherwise specified. Validation and test reference captions are not used to construct the training knowledge base.

5.2 Visual overview of the exploratory analysis

The figures in this chapter complement the numerical tables by showing the distributions and relationships observed in the training subset. They are loaded from the

`figures/` directory in the Overleaf project. The source below uses descriptive filenames; if the uploaded files have different names, only the filename in the corresponding `\projectfigure` command needs to be changed.

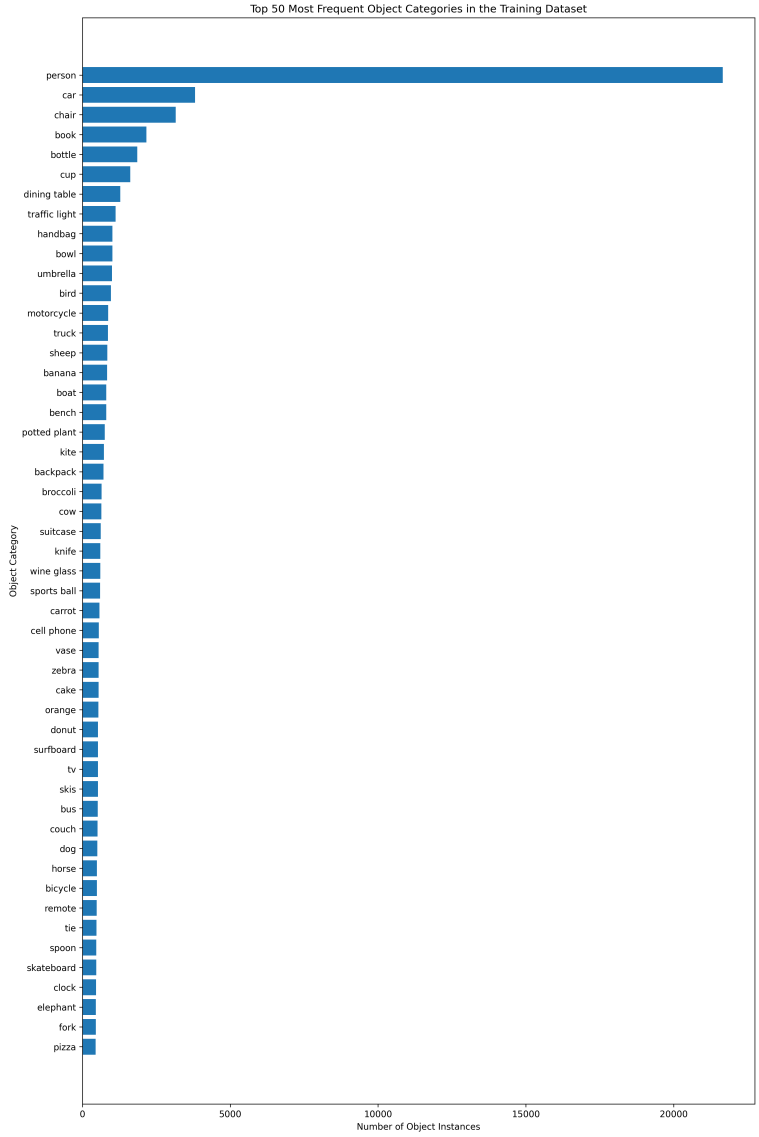


Figure 5.1: Frequency of the 50 most common object classes in the training subset.

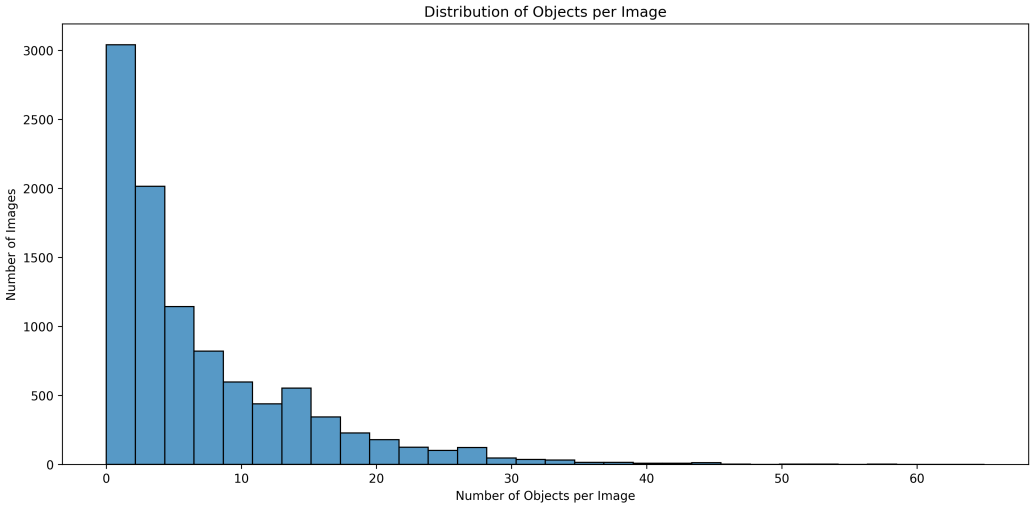


Figure 5.2: Distribution of annotated objects per image, illustrating scene complexity.

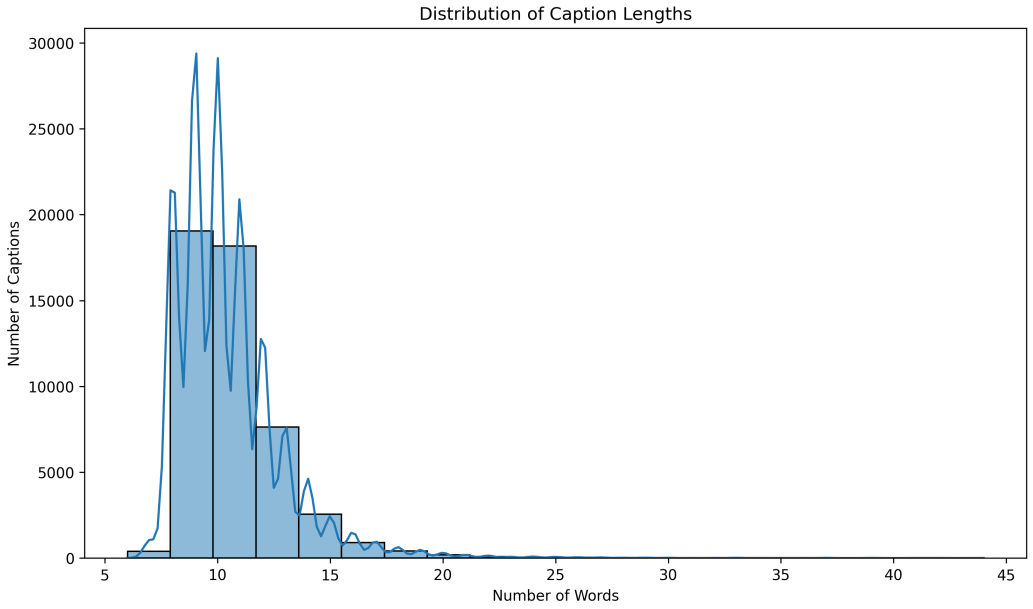


Figure 5.3: Distribution of human-written caption lengths.

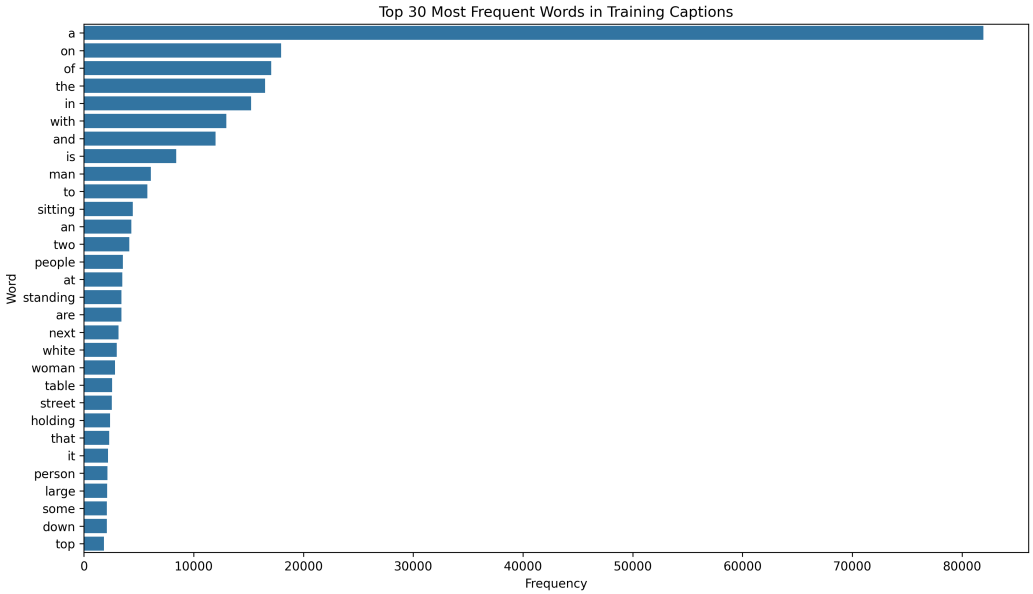


Figure 5.4: The 30 most frequent words in the human-written captions.

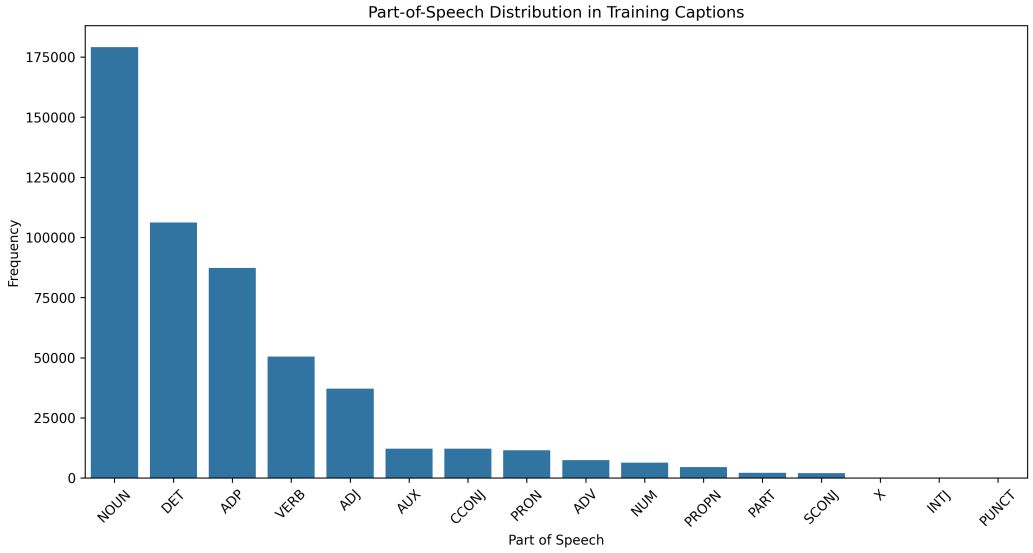


Figure 5.5: Part-of-Speech distribution across the caption corpus.

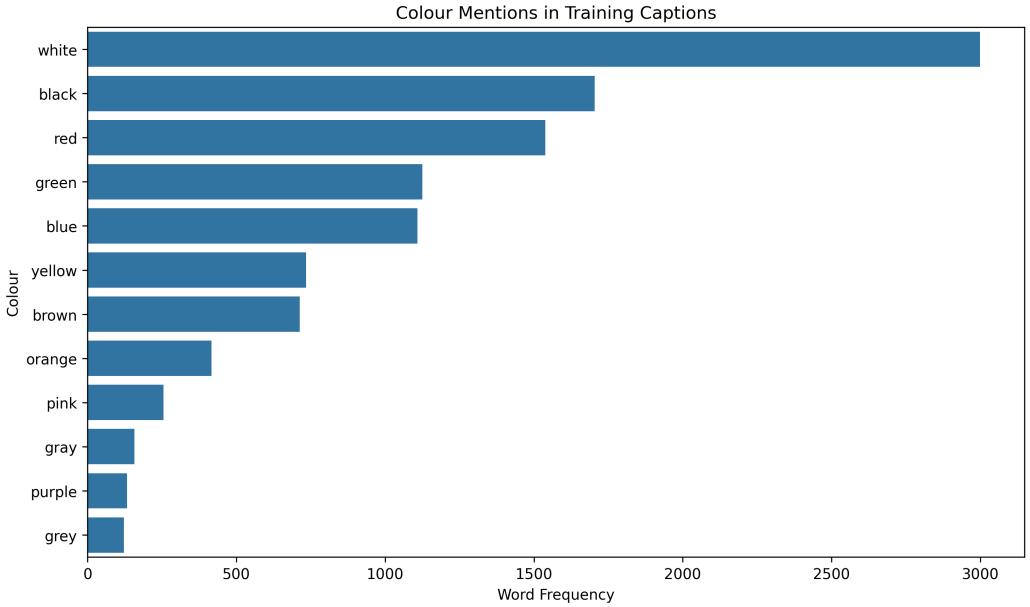


Figure 5.6: Frequency of colour-related mentions in the captions.

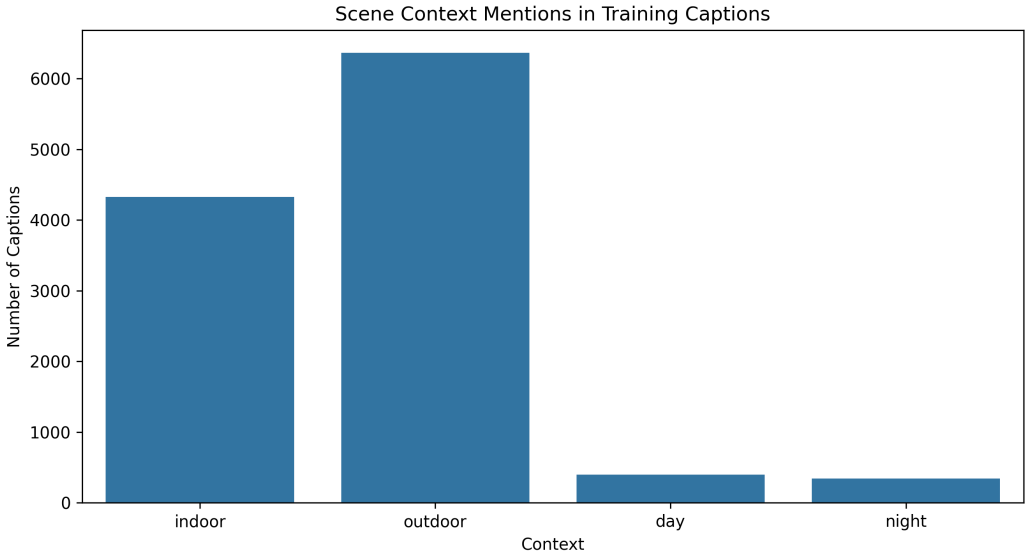


Figure 5.7: Frequency of context-related mentions in the captions.

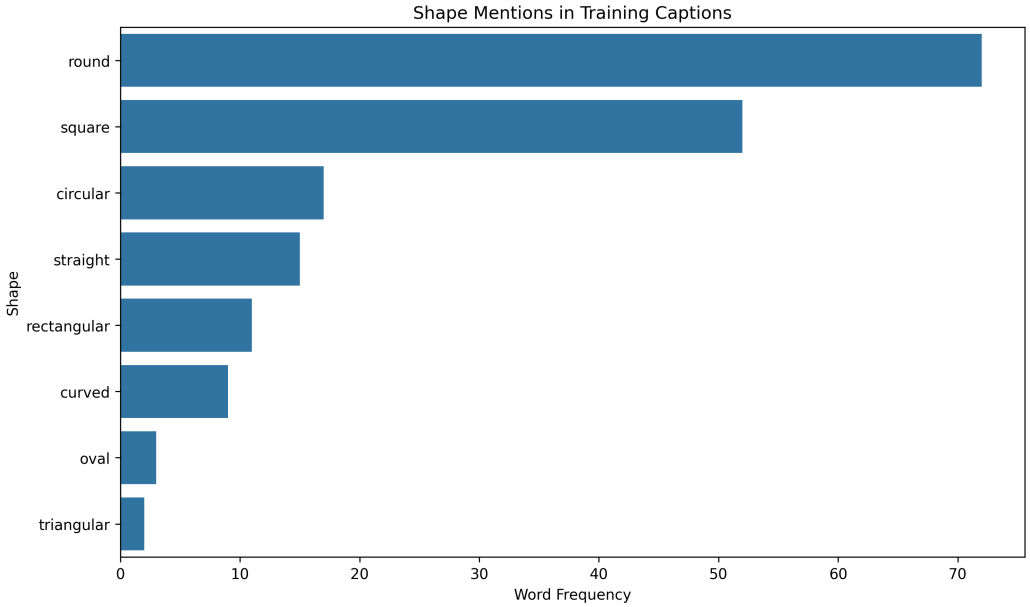


Figure 5.8: Frequency of shape-related mentions in the captions.

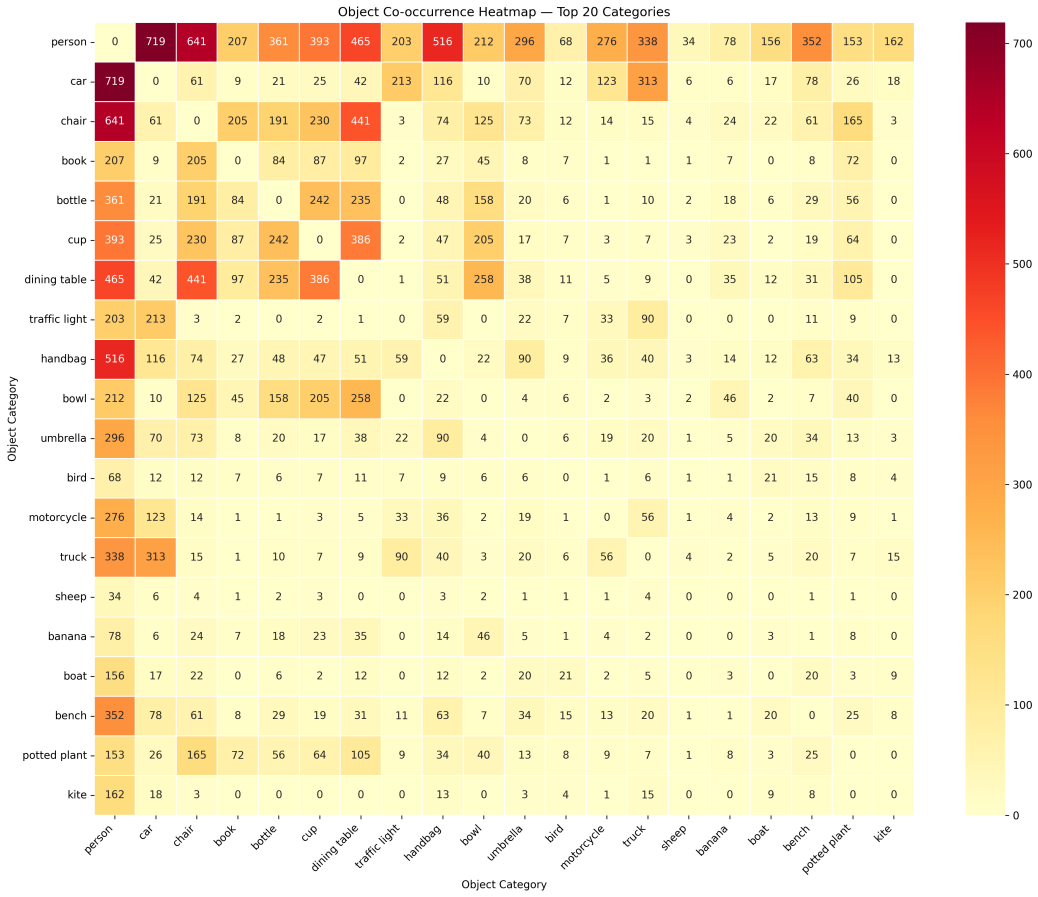


Figure 5.9: Heatmap of object-category co-occurrence, highlighting frequently shared image contexts.

5.3 Structured data inspection

COCO annotation files are JSON documents. For analysis, caption and instance annotations are loaded into structured Python objects and converted into Pandas DataFrames.

A simplified example of loading caption annotations is shown below:

```
import json
import pandas as pd

with open("captions_train2017.json",
          "r", encoding="utf-8") as file:
    captions_data = json.load(file)

captions_df = pd.DataFrame(
    captions_data["annotations"]
)

print(captions_df.head())
```

A tabular representation makes it possible to group captions by image identifier, count annotations, calculate caption lengths, inspect missing values, and join caption information with image metadata.

The same approach is applied to instance annotations to examine object categories and the number of annotated instances per image.

5.4 Dataset scale and general statistics

The selected training subset contains 9,900 images, 49,500 captions, and 71,326 object annotations. The vocabulary extracted from the captions contains 9,514 distinct words.

Table 5.1: Main descriptive statistics of the training subset.

Measure	Value
Training images	9,900
Training captions	49,500
Captions per image	5
Object annotations	71,326
Distinct COCO object categories	80
Distinct normalized words	9,514
Total word occurrences	518,774

5.5 Object-category frequency

The first analysis examines the frequency of the object categories defined by COCO. The category identifiers in the instance annotations are mapped to category names, and the number of occurrences of each category is counted.

The ten most frequent categories reported in the exploration are shown in Table 5.2.

Table 5.2: Ten most frequent object categories.

Category	Number of annotations
Person	21,658
Car	3,807
Chair	3,153
Book	2,166
Bottle	1,858
Cup	1,622
Dining table	1,278
Traffic light	1,123
Handbag	1,013
Bowl	1,012

The person category accounts for 21,658 annotations, approximately 30.36% of all object annotations. It is substantially more frequent than the other categories. Car and chair are the next most frequent categories.

This imbalance is relevant when interpreting the contents of the knowledge base. Common categories have more examples available than rare categories. Consequently, later retrieval experiments should be interpreted with awareness of the distribution of

object categories in the underlying dataset.

5.6 Scene complexity

Scene complexity is approximated by counting the number of COCO object annotations associated with each image. The resulting statistics are presented in Table 5.3.

Table 5.3: Scene-complexity statistics.

Measure	**Value**
Number of images	9,900
Total object annotations	71,326
Mean objects per image	7.20
Median objects per image	4
Standard deviation	7.27
Minimum objects per image	0
Maximum objects per image	65

The mean of 7.20 is higher than the median of four. This difference indicates that images with a large number of annotated objects increase the mean. The dataset therefore contains both relatively simple images and complex scenes with many annotated instances.

The reported counts are based on the COCO instance annotations. They should not be interpreted as a complete count of every object visually present in an image, since not every visible entity is necessarily annotated.

5.7 Caption-length analysis

Caption length is measured by counting the words in each human-written description. The analysis covers all 49,500 training captions.

Table 5.4: Caption-length statistics.

Measure	**Value**
Number of captions	49,500
Mean length	10.46 words
Median length	10 words
Standard deviation	2.39 words
Minimum length	6 words
Maximum length	44 words

The results show that captions are generally concise, with an average length of approximately ten words. The maximum length of 44 words indicates that a smaller number of captions are substantially longer than the typical description.

This finding supports preserving complete captions as textual records during the current preparation stage. The captions are already relatively short, and splitting them without a specific experimental reason could separate words that contribute to a single coherent description.

5.8 Vocabulary analysis

The vocabulary analysis measures the lexical diversity of the training captions. The text is normalized to lowercase, and alphabetic words are extracted for counting.

The analysis reports 518,774 word occurrences and 9,514 distinct words. The most frequent word is `a`, with 81,950 occurrences. Other frequent words include grammatical terms such as *on*, *of*, *the*, *in*, and *with*, as well as visually meaningful terms such as *man*, *sitting*, *people*, *standing*, *woman*, and *table*.

The results show that the captions combine grammatical words, which provide sentence structure, with nouns and verbs that convey visual content. This is one reason to avoid overly aggressive text cleaning that could remove information needed to interpret the captions.

5.9 Part-of-Speech distribution

Part-of-Speech (POS) tagging was performed using spaCy and the English model `en_core_web_sm`. The purpose was to examine the grammatical composition of the captions.

Table 5.5: Principal Part-of-Speech categories.

POS category	**Frequency**	**Percentage**
Nouns	179,165	34.53%
Determiners	106,217	20.47%
Adpositions	87,364	16.84%
Verbs	50,435	9.72%
Adjectives	37,191	7.17%
Auxiliary verbs	12,222	2.36%
Coordinating conjunctions	12,178	2.35%
Pronouns	11,551	2.23%
Adverbs	7,420	1.43%

Nouns form the largest category, accounting for 34.53% of the analysed tokens. This is consistent with the role of captions in identifying people, animals, and objects. Verbs describe actions, while adjectives contribute descriptive information about visual attributes.

Nouns, verbs, and adjectives are therefore particularly relevant to image description: they identify entities, express activities, and provide additional details.

5.10 Visual attribute and contextual diversity

The exploration also examines how frequently captions mention colours, shapes, and scene contexts. These counts are based on predefined vocabularies and therefore measure textual mentions rather than independently verified visual properties.

5.10.1 Colour terms

Table 5.6: Selected colour terms in the training captions.

Colour	Word occurrences	Captions mentioning colour
White	2,999	2,894
Black	1,704	1,675
Red	1,538	1,513
Green	1,125	1,109
Blue	1,109	1,084
Yellow	734	726
Brown	713	701
Orange	417	410

White is the most frequently mentioned colour, followed by black and red. The results show that colour information occurs in the human-written captions, but not uniformly across the dataset.

5.10.2 Shape terms

Table 5.7: Shape-related word occurrences.

Shape term	Occurrences
Round	72
Square	52
Circular	17
Straight	15
Rectangular	11
Curved	9
Oval	3
Triangular	2

Shape-related words are much less frequent than colour-related words. This indicates that the captions in the selected subset more often mention colours than explicit geometric descriptions.

5.10.3 Scene-context terms

Table 5.8: Selected scene-context mentions.

Context	Word occurrences	Captions mentioning context
Indoor	4,551	4,328
Outdoor	6,557	6,365
Day	516	397
Night	350	343

Outdoor-related terms occur more frequently than indoor-related terms. Explicit references to daytime and nighttime are less common. These values describe the language used in the captions and should not be treated as definitive scene classifications.

5.11 Object-category co-occurrence

The co-occurrence analysis examines which object categories appear together in the same image. For each image, the distinct annotated categories are extracted, and category pairs are counted across the training subset.

Table 5.9: Most frequent object-category co-occurrences.

Category pair	Number of images
Person – Car	719
Person – Chair	641
Person – Handbag	516
Person – Dining table	465
Chair – Dining table	441
Person – Backpack	405
Person – Cup	393
Cup – Dining table	386
Person – Bottle	361
Person – Bench	352

The person category appears frequently with several other categories. Furniture and dining-related objects also show recurring combinations, as illustrated by the chair–dining table and cup–dining table pairs.

Co-occurrence indicates that categories are present in the same image; it does not, by itself, establish a spatial relationship or an interaction between the objects.

The corresponding co-occurrence heatmap provides a matrix-based view of these relationships. Each cell represents the number of images in which a pair of categories occurs together (according to the project's co-occurrence calculation). Stronger cells indicate category pairs that are more frequently observed in the same images. The heatmap describes co-presence only; it should not be interpreted as evidence of a causal, spatial, or semantic relationship.

5.12 Examples of image-caption relationships

The knowledge-base inspection and retrieval compatibility notebook provide examples of images and their associated human-written captions. These examples illustrate the variation in wording and the importance of preserving the correct image-caption association.

Example 1: Traffic signs and street infrastructure

The record associated with image ID `11605` includes captions such as:

> "A couple of one way street signs hanging on a traffic light."

> "Two one way signs are on the same pole as a stop light."

The captions describe related visual content using different wording. They refer to street signs, a pole, and traffic infrastructure.

Example 2: Street-name signs

The record associated with image ID `401670` includes captions such as:

> "A pole holding a couple of street signs beside a building."

> "A view of street signs on a pole."

These descriptions show that captions associated with the same image may differ in their level of detail while remaining focused on the same principal visual elements.

Example 3: A street sign and its surroundings

The record associated with image ID `486769` includes captions such as:

> "A street sign of West 76 street near a tree."

> "The street sign says W 76 St with a blue sky in the background."

In this example, one caption emphasizes the street sign and nearby tree, while another mentions the background sky. The examples demonstrate the descriptive variation available in the five human-written captions associated with each image.

5.13 Consequences for data preparation

The exploration results informed several decisions in the preparation stage.

- **Category imbalance:** the high frequency of the person category should be considered when interpreting the coverage of the training collection.
- **Variable scene complexity:** images contain different numbers of annotated objects, so image identifiers and associated annotations must be preserved accurately.
- **Concise captions:** the average caption length supports retaining complete captions as individual textual records.
- **Vocabulary diversity:** semantic differences in wording motivate the use of sentence embeddings rather than relying only on exact word matching.
- **Caption variation:** retaining all five captions per image preserves the different descriptions provided by human annotators.
- **Object co-occurrence:** category-pair statistics provide additional information about the visual composition of the selected subset.

Overall, the exploration phase provided a quantitative description of the data and helped establish appropriate preparation and validation requirements.

Chapter 6

Preprocessing and Quality Control

6.1 Preprocessing objectives

The purpose of preprocessing is to ensure that the selected data can be represented consistently and used reliably in subsequent project activities. The preparation process must preserve the original relationships between images, captions, and object annotations.

The current preparation workflow includes the inspection of source records, the handling of textual representation, the generation of embeddings, and the organization of metadata. The project uses a conservative approach to caption preparation so that useful linguistic information is not removed unnecessarily.

6.2 Embedding integrity

The generated embedding arrays were checked for their expected dimensions, numerical validity, and consistency with the corresponding document collections.

Table 6.1: Embedding integrity checks.

Validation	**CLIP image embeddings**	**SBERT caption embeddings**
Expected vectors	9,900	49,500
Actual vectors	9,900	49,500
Dimensions	512	384
NaN values	0	0
Infinite values	0	0
Zero vectors	0	0
Data type	`float32`	`float32`
L2 normalization	Passed	Passed

The checks confirm that the generated embedding collections contain the expected number of vectors and that no NaN, infinite, or zero vectors were reported. The stored vectors also satisfy the configured data-type and normalization requirements.

6.3 Identifier and document consistency

The quality-control process checks the relationships between image identifiers, caption identifiers, document records, and embedding rows.

The checks include:

- Uniqueness of image identifiers.
- Uniqueness of caption identifiers.
- Alignment between embedding rows and document records.
- Consistency between image documents and their associated caption identifiers.
- Existence of every caption referenced by an image document.
- Agreement between the image identifier in a caption document and the identifier of its associated image.

The quality-control report records 9,900 image documents and 49,500 caption documents. It also records 49,500 caption references, all unique, with no unreferenced captions or duplicate caption references.

Table 6.2: Document and identifier consistency results.

Check or quantity	**Result**
Image documents	9,900
Caption documents	49,500
Caption references	49,500
Unique caption references	49,500
Unreferenced captions	0
Duplicate caption references	0
Mapping uniqueness and row alignment	Passed
Embedding-to-document consistency	Passed
Image-caption relationships	Passed

These results are important because a mismatch between an image and its caption could introduce incorrect information into any later retrieval-supported process.

6.4 Leakage audit

Data leakage occurs when information belonging to one evaluation subset becomes accessible in another subset in a way that compromises the validity of the experiment. In this project, the knowledge base must contain training images and their associated training captions only.

The validation and test subsets are kept separate from the training knowledge base. The leakage audit checks the intersections between the image-ID sets and verifies that no validation or test image is included in the knowledge base.

Table 6.3: Image-ID leakage audit.

Leakage check	**Overlapping IDs**
Training – Validation	0
Training – Test	0
Validation – Test	0
Validation images in knowledge base	0
Test images in knowledge base	0
Non-training images in knowledge base	0

The reported audit found no overlapping identifiers between the three subsets. It also confirmed that the knowledge base contains no validation or test images. This separation is necessary to avoid using evaluation examples as part of the information collection against which they will later be assessed.

6.5 FAISS index validation

The two FAISS indexes were checked against their corresponding embedding matrices.

Table 6.4: Vector-index validation.

Property	**Image index**	**Caption index**
Index type	IndexFlatIP	IndexFlatIP
Number of vectors	9,900	49,500
Vector dimensions	512	384
Similarity operation	Inner product	Inner product
Vector count consistency	Passed	Passed

The index dimensions and vector counts correspond to the generated embedding

collections. Since the vectors are L2-normalized, inner-product search provides cosine-similarity ranking within each index.

6.6 Statistical output verification

The statistical exploration produced report artifacts in the form of figures, JSON summaries, and CSV tables. The repository quality-control documentation records the following generated outputs:

Table 6.5: Statistical analysis artifacts.

Artifact type	**Quantity**
Statistical figures	5
Statistical JSON files	5
CSV tables	4

These artifacts support inspection and reproducibility of the descriptive analysis. They also provide a record of the statistics used to characterize the training subset.

6.7 Retrieval-index compatibility test

As a final technical check of the prepared data artifacts, the project executed an image-retrieval compatibility test. This test was performed using a validation image as a query against the training image index.

The test configuration is summarized in Table 6.6.

Table 6.6: Image-retrieval compatibility test configuration.

Parameter	**Value**
Query image ID	250
Query split	Validation
Image encoder	CLIP ViT-B/32
Embedding dimension	512
Index	FAISS IndexFlatIP
Top-k	3
Similarity measure	Inner product

The query image was encoded using CLIP and its embedding was searched against the training image index. The three nearest training images were then mapped to their

document records, and their associated caption records were loaded.

Table 6.7: Nearest training images returned by the compatibility test.

Rank	Training image ID	Similarity score
1	11,605	0.8479
2	401,670	0.8389
3	486,769	0.8303

The three retrieved images were associated with fifteen captions in total. The notebook verified that the caption records could be loaded and that their image identifiers matched the corresponding retrieved image records.

This test demonstrates technical compatibility between the CLIP encoder, image embedding collection, FAISS index, image documents, and caption documents. It is a smoke test of the prepared retrieval artifacts, not a complete evaluation of retrieval relevance or caption-generation quality.

6.8 Phase 1 status

The principal outcomes of Phase 1 are summarized below.

Table 6.8: Summary of Phase 1 activities and status.

Activity	Status
Dataset inspection and annotation loading	Completed
Reproducible dataset splitting	Completed
Object-category analysis	Completed
Scene-complexity analysis	Completed
Caption-length analysis	Completed
Vocabulary analysis	Completed
Part-of-Speech analysis	Completed
Colour, shape, and context analysis	Completed
Object co-occurrence analysis	Completed
Image embedding generation	Completed
Caption embedding generation	Completed
Knowledge-base document organization	Completed
Embedding integrity checks	Passed
Identifier and document consistency checks	Passed
Leakage audit	Passed

Activity	**Status**
FAISS index validation	Passed
Retrieval compatibility smoke test	Passed

Chapter 7

Conclusion

7.1 Summary of achievements

Phase 1 established the dataset and knowledge-base foundations for the project *Image Captioning using Retrieval-Augmented Generation*. The work began with the inspection and organization of the Microsoft COCO 2017 dataset and the definition of reproducible training, validation, and test subsets.

A detailed statistical exploration was conducted on the training subset. It examined object-category frequency, scene complexity, caption length, vocabulary size, Part-of-Speech distribution, visual attribute mentions, scene-context terms, and object-category co-occurrence. The results provided a quantitative understanding of the visual and linguistic characteristics of the selected data.

The project then generated 9,900 image embeddings using CLIP and 49,500 caption embeddings using Sentence-BERT. The two collections were maintained separately and organized into independent FAISS indexes. Structured image and caption documents were created to preserve the associations between images, captions, and embedding rows.

A quality-control process verified the numerical integrity of the embeddings, the uniqueness of identifiers, the consistency of document mappings, and the correspondence between images and captions. The leakage audit found no overlap between the training, validation, and test image identifiers and confirmed that the knowledge base contains training data only.

Finally, an image-retrieval compatibility test demonstrated that the prepared image index and document mappings could be loaded and used to retrieve training examples and their associated captions.

These achievements provide a structured and validated data foundation for the project. They represent the outcomes of Phase 1 and should not be interpreted as results from a completed caption-generation evaluation.

7.2 Phase 1 conclusion

The main objective of Phase 1 was to prepare a reliable, organized, and reproducible dataset that can support the subsequent development and evaluation of the project. Based on the implemented preprocessing, embedding generation, knowledge-base organization, and quality-control procedures, this objective has been substantially achieved.

The most important outcome is not simply the creation of vector indexes, but the establishment of consistent relationships between the original images, their human-written captions, their annotations, and their numerical representations. This consistency is essential for ensuring that subsequent experiments are conducted on a controlled and well-documented data foundation.

The project can therefore proceed from the Phase 1 data-preparation results, while preserving the split definitions and quality-control evidence established in this report.

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